

UNIT-4

Software Quality Models

ISO 9126

- ISO 9126 is an international standard for software quality evaluation that was fully adopted in 1992.
- It defines a comprehensive framework for measuring software quality through six main characteristics:
- **Functionality:** The degree to which the software meets specified requirements.
- **Reliability:** The ability of the software to perform under specified conditions for a specified period.
- **Usability:** The ease with which users can learn and operate the software.
- **Efficiency:** The performance of the software in terms of resource usage.
- **Maintainability:** The ease with which the software can be modified.
- **Portability:** The ability of the software to be transferred from one environment to another.

ISO/IEC 25010

- The ISO/IEC 25010 standard outlines a comprehensive framework for assessing software quality.
- It consists of two primary models:
 - 1) Product Quality Model: This model includes eight characteristics:
 - Functional Suitability
 - Reliability
 - Performance Efficiency
 - Usability
 - Security
 - Compatibility
 - Maintainability
 - Portability
 - 2) Quality in Use Model: Comprising five characteristics that focus on the effectiveness and satisfaction of users when interacting with the software

McCall's Model

- Developed by Jim McCall in 1977, McCall's Quality Model categorizes software quality into three main areas:
- Product Operation:
 - Includes factors such as correctness, reliability, efficiency, integrity, and usability.
- Product Revision:
 - Focuses on maintainability, flexibility, and testability.
- Product Transition:
 - Addresses portability, reusability, and interoperability.
- McCall's model distinguishes between two levels of quality attributes:
- Quality Factors: High-level attributes that can be measured directly.
- Quality Criteria: Lower-level attributes that may be assessed subjectively or objectively.

Boehm's Model

- Introduced by Barry Boehm in 1978, this model emphasizes a hierarchical structure of software quality attributes. It defines seven key characteristics-
- Utility
- Reliability
- Efficiency
- Human Engineering
- Testability
- Maintainability
- Portability

Boehm's Model

- Additionally, it categorizes software systems into three quality levels:
- As-is Utility
- Maintainability
- General Utility
- Boehm's model is particularly useful for assessing complex systems where multiple attributes need consideration simultaneously.
- It allows stakeholders to evaluate trade-offs between different quality attributes during development.

Capability Maturity Model (CMM)

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CMM Maturity Levels

- CMM consists of five distinct maturity levels, each representing a different stage in the evolution of an organization's software processes:
- Initial
- Repeatable
- Defined
- Managed
- Optimizing

Initial (Level 1)

- Characteristics: Processes are typically undocumented and chaotic. Success relies heavily on individual efforts rather than established practices.
- Focus: Organizations at this level often lack any formalized processes, leading to unpredictable outcomes and inefficiencies

Repeatable (Level 2)

- Characteristics: Basic project management processes are established. Work is planned, tracked, and can be repeated for similar projects.
- Focus: Organizations begin to implement standard practices for managing costs, schedules, and functionalities, allowing them to build on previous successes

Defined (Level 3)

- Characteristics: Processes are well-defined and documented across the organization. There is a common understanding of roles and responsibilities.
- Focus: Emphasis is placed on standardizing processes to ensure consistency in project management and software development activities

Managed(Level 4)

- Characteristics: Organizations use quantitative techniques to manage processes effectively. Performance metrics are established to monitor progress.
- Focus: This level emphasizes quality management through precise measurements, allowing organizations to predict process performance reliably

Optimizing(Level 5)

- Characteristics: Continuous process improvement is the hallmark of this level. Organizations are agile and can adapt processes based on performance data.
- Focus: At this stage, organizations focus on innovation and proactive improvements, ensuring that processes are not only efficient but also capable of evolving with changing needs

Key Process Areas (KPAs)

- Each maturity level, except for Level 1, is associated with specific Key Process Areas (KPAs) that organizations should focus on to advance their maturity:
- Level 2 KPAs: Requirements Management, Project Planning, Project Monitoring and Control, Supplier Agreement Management, Measurement and Analysis.
- Level 3 KPAs: Organization Process Focus, Organization Process Definition, Training Program, Integrated Software Management, Software Product Engineering, Inter-group Coordination, Peer Reviews.
- Level 4 KPAs: Quantitative Process Management, Software Quality Management.
- Level 5 KPAs: Organizational Innovation and Deployment, Causal Analysis and Resolution.

Advantages of CMM

- Resource Optimization: Helps organizations utilize resources more effectively by identifying wasteful practices.
- Benchmarking and Evaluation: Provides a formal structure for self-assessment against industry standards.
- Quality Management: Encourages the implementation of best practices for quality assurance.
- Process Enhancement: Offers a systematic approach for continuous improvement in organizational processes.

Challenges in CMM

- Resistance to Change
- Lack of Leadership Support
- Resource Constraint
- Organizational Culture Misalignment
- Complexity of CMMI
- Lack of Expertise
- Integration Challenges
- Focus on Certification Over Improvement